



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 1 Introduction to general pathology and biopsy

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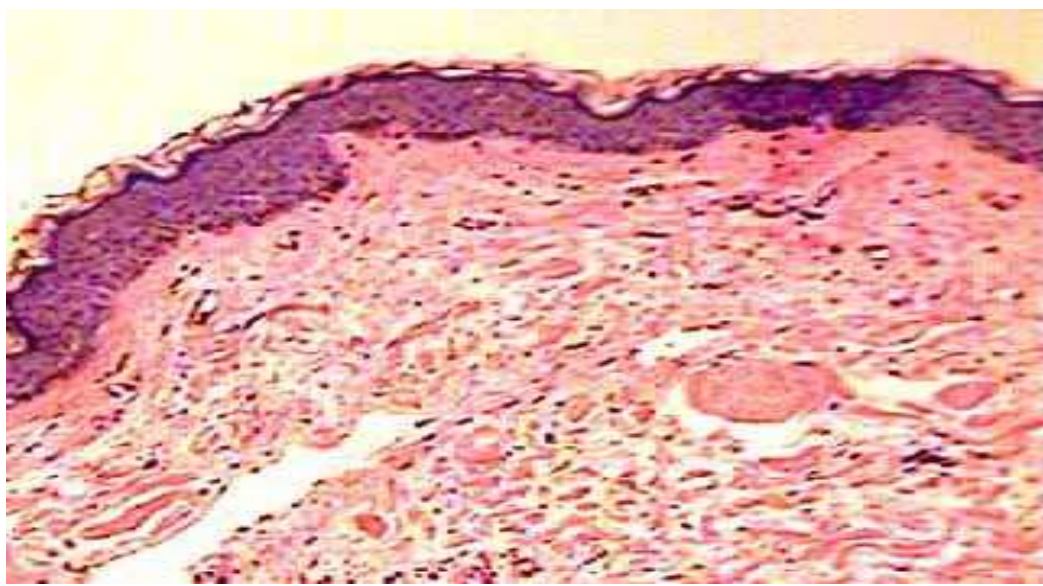
Introduction

Pathology: derived from the Greek word's pathos (suffering or disease) and logos (study), is the scientific specialty that studies the nature of disease. It acts as the bridge between basic sciences and clinical medicine.

We can broadly divide pathology into two areas:

A-General Pathology: It's the study of the basic, universal reactions of cells and tissues to injury. These fundamental processes include **inflammation, cell death (necrosis and apoptosis), tissue repair, and cellular adaptation to stress.**

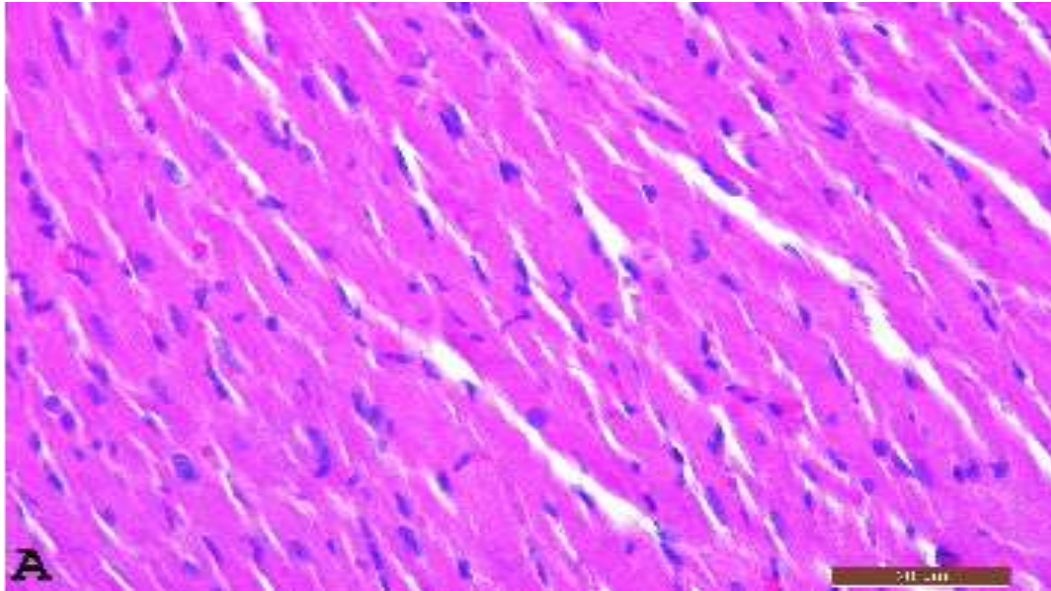
B-Systemic Pathology: This branch applies the principles of general pathology to specific organs and organ systems, like **cardiovascular pathology, oral pathology.**



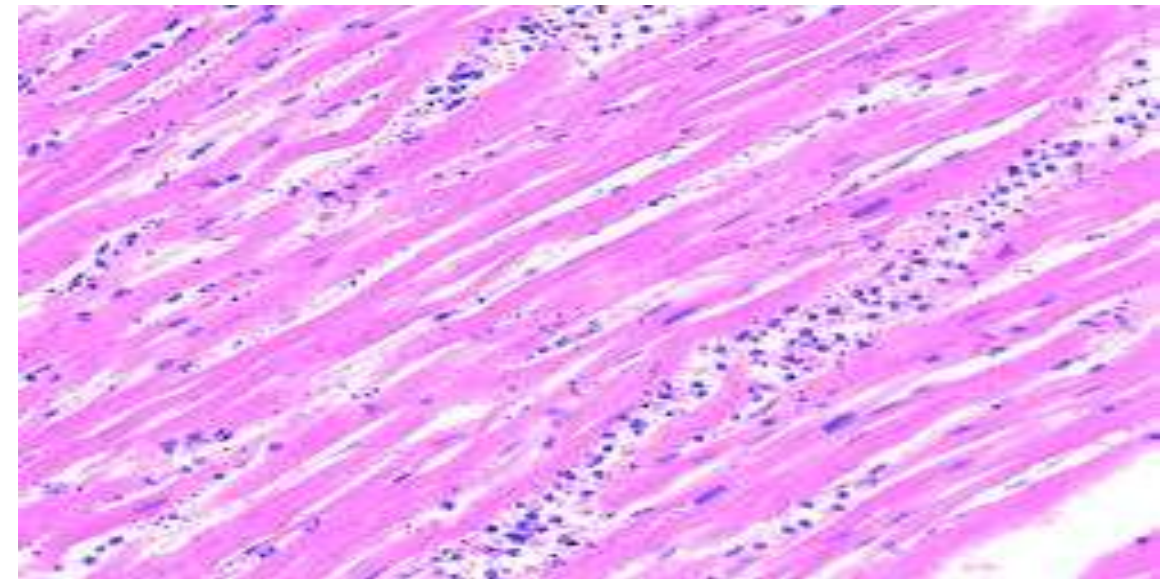
Normal Skin.



**Skin Changes in response to an insect bite.
(General Pathology , Inflammation)**



Normal Cardiac Tissue.



**Cardiac Tissue Changes in response to reduced blood supply.
(Systematic Pathology , Myocardial Infarction)**

Obtaining Tissue Sample

The most common procedure used in histologic research is a biopsy which involves the extraction of a small tissue sample from an organ or specific body area, intended for close examination using a microscope.

Some examples of biopsies;

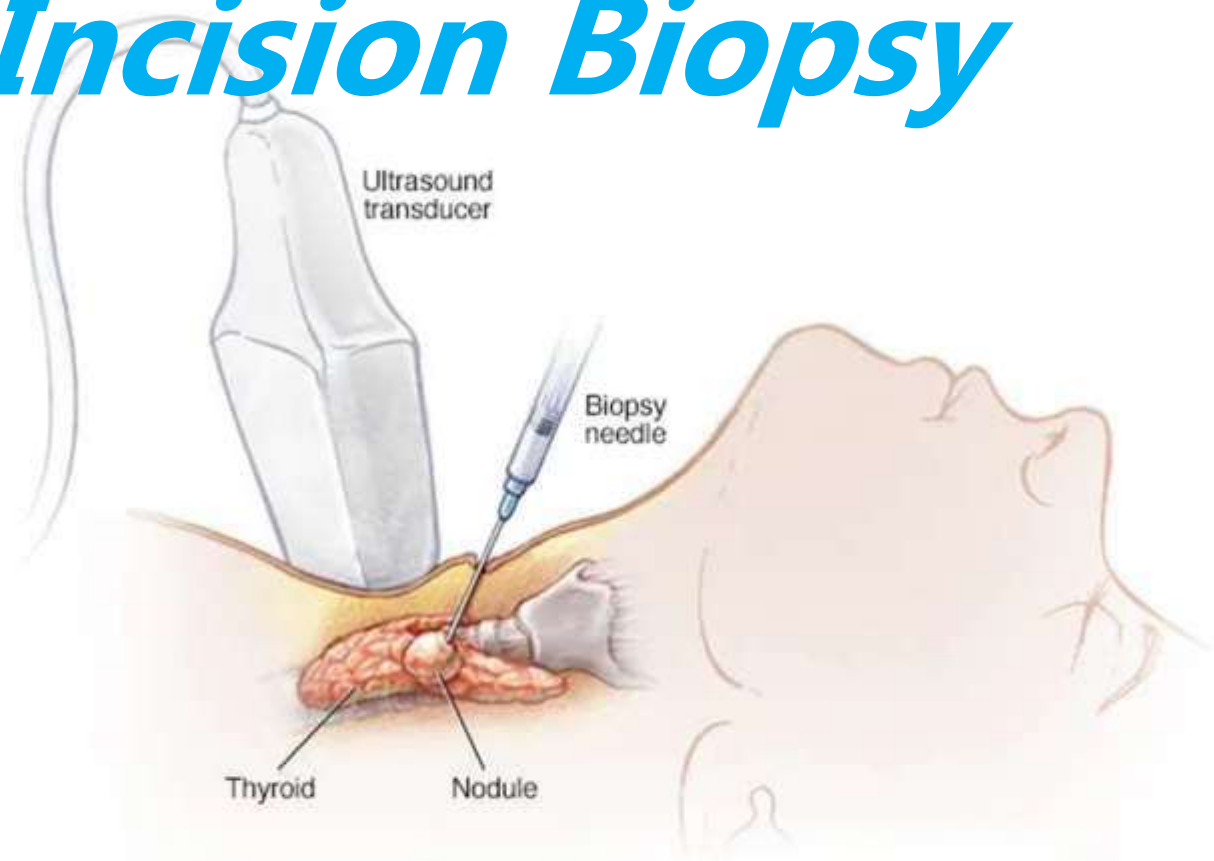
1-Needle Biopsy: Applied to areas like the brain, breast, liver, kidney, and muscle.

2-Direct Incision Biopsy: Used on surfaces like the skin, mouth, and larynx.

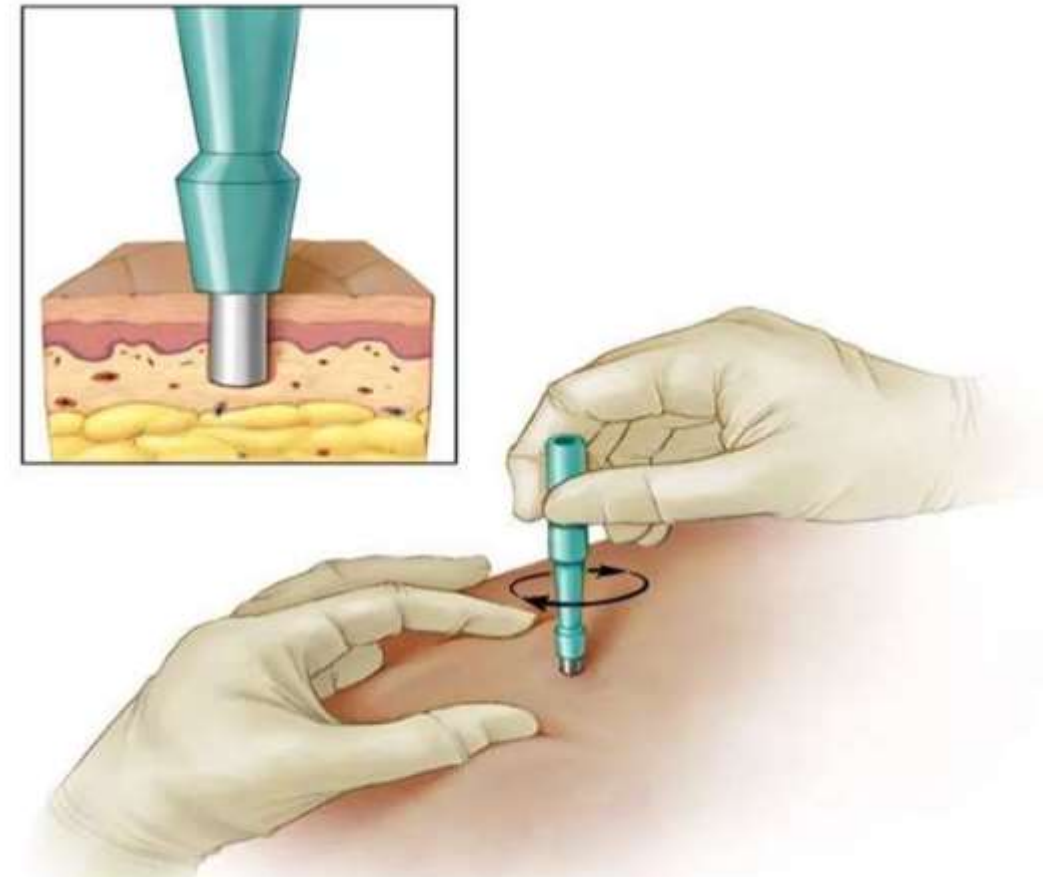
3-Endoscopic Biopsy: Employed in regions such as the lung, intestine, and bladder.

4-Transvascular Biopsy: Used in like the heart and liver.

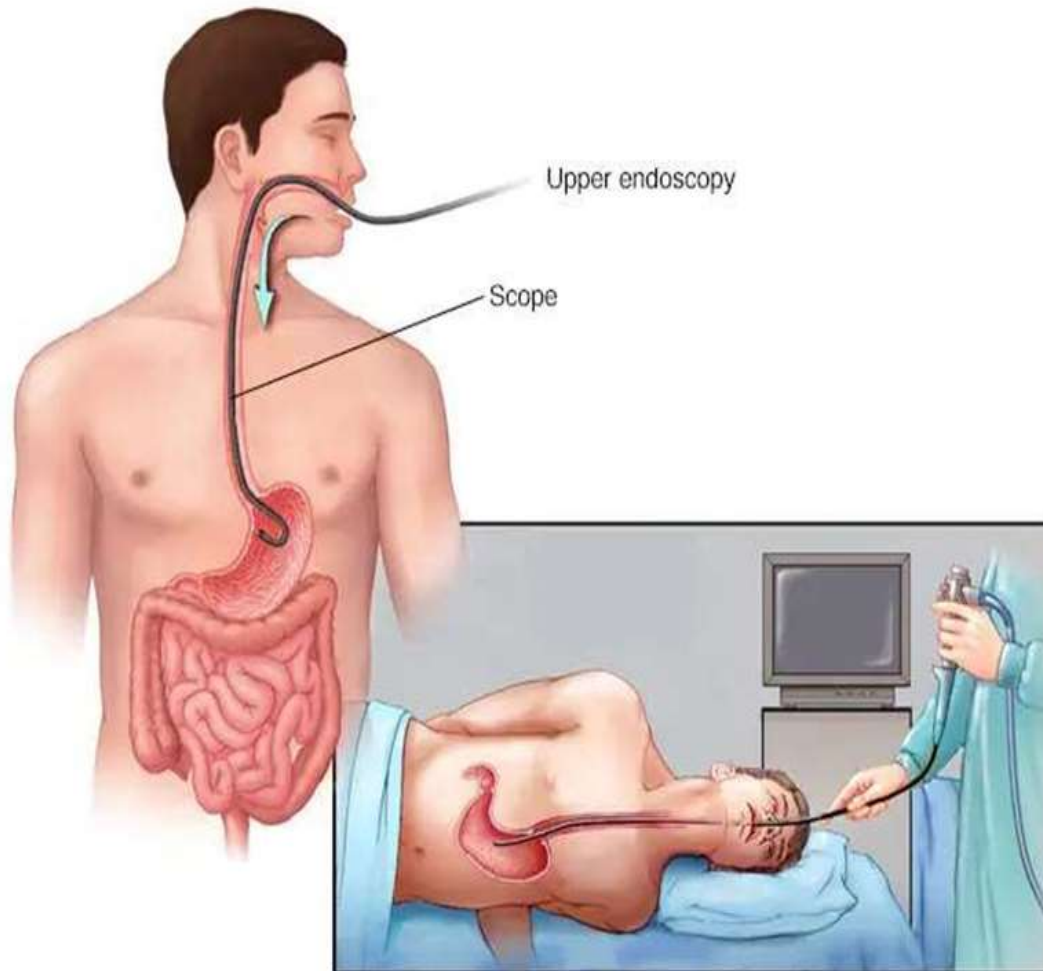
Needle Biopsy ***Incision Biopsy***



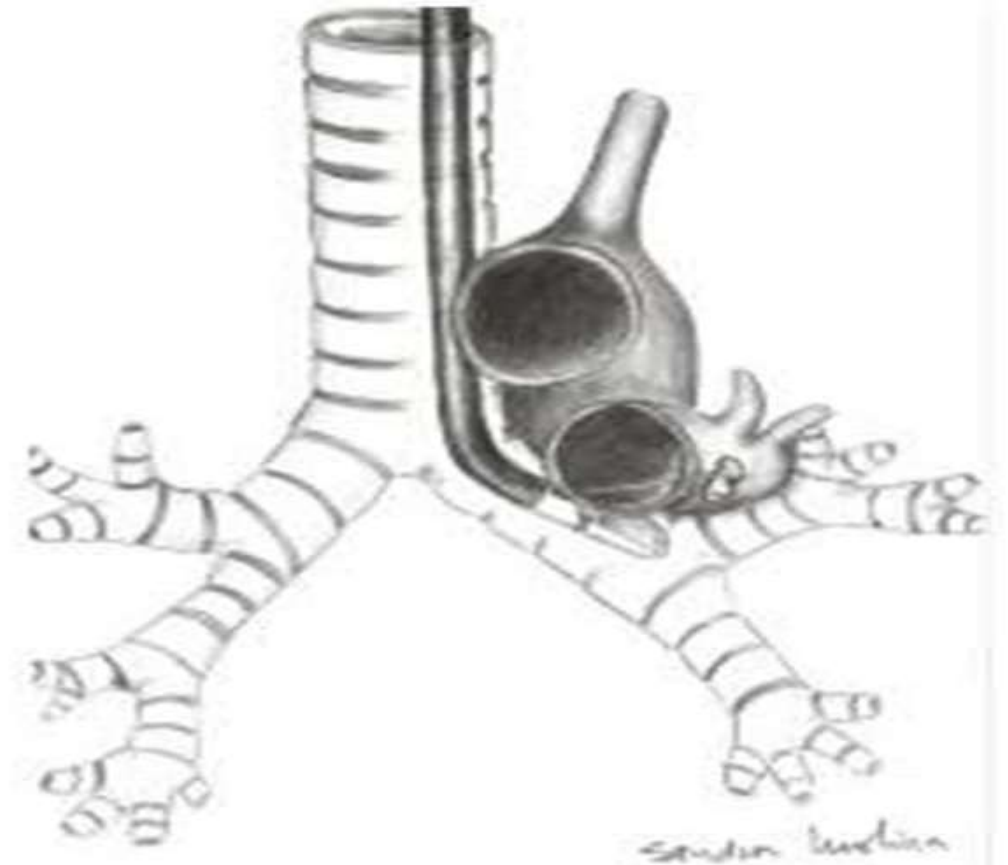
Direct



Endoscopic Biopsy



Transvascular



Steps in tissues preparation

1-Fixation:

Purpose: To preserve the cellular structure and prevent decay or autolysis (self- destruction of cells).

Method: Tissue samples are placed in a fixative solution, often formaldehyde, that chemically stabilizes the proteins and cellular components.

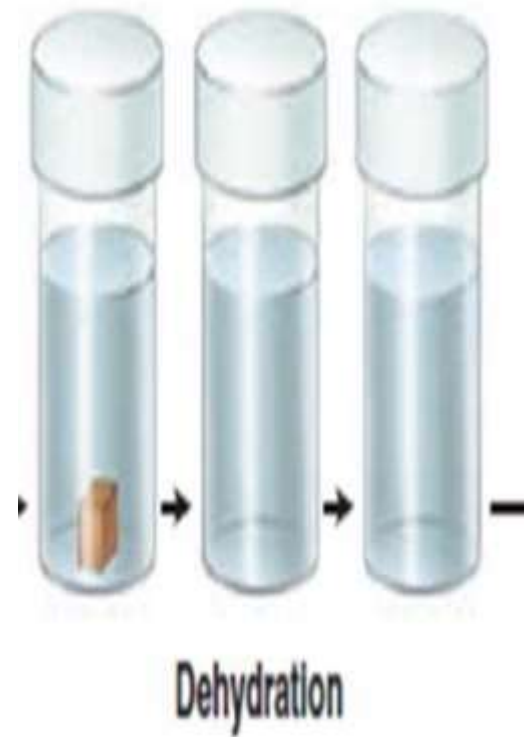


Steps in tissues preparation

2-Dehydration:

Purpose: To remove water from the tissue, which is essential for embedding in solid mediums.

Method: Tissue is treated with a series of increasing concentrations of alcohol (ethanol) to gradually replace water with alcohol.

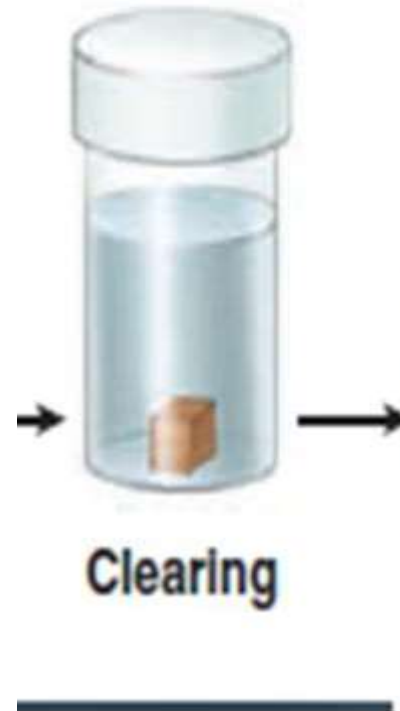


Steps in tissues preparation

3-Clearing:

Purpose: To make the tissue transparent, allowing better infiltration of the embedding medium.

Method: Tissue is immersed in a clearing agent (xylene) that dissolves the alcohol while maintaining tissue structure.

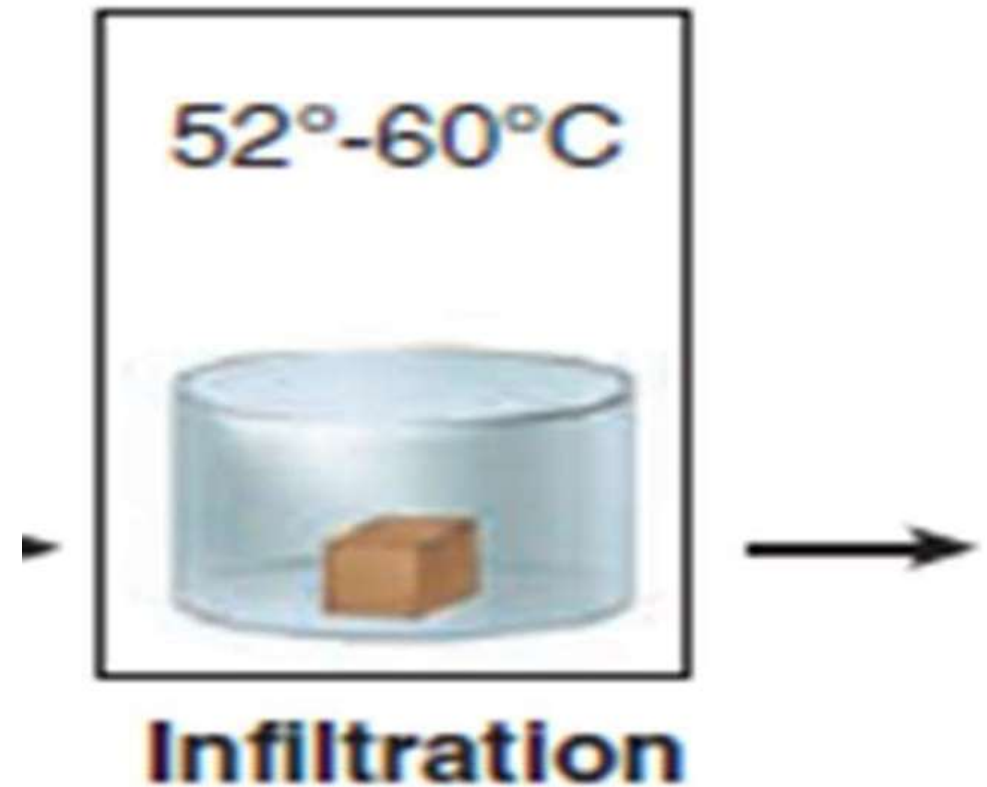


Steps in tissues preparation

4-Infiltration:

Purpose: To replace the clearing agent with an embedding medium, providing structural support for tissue sectioning.

Method: The fully cleared tissue is placed in melted paraffin wax in an oven at 52°-60°C.

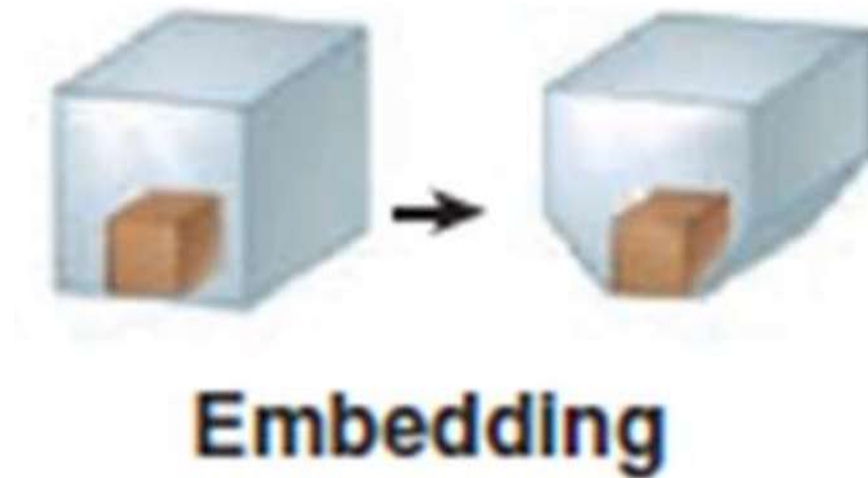


Steps in tissues preparation

5-Embedding:

Purpose: To encase the tissue in a solid block that can be securely held for sectioning.

Method: The paraffin-infiltrated tissue is placed in a small mold with melted paraffin and allowed to harden.



Steps in tissues preparation

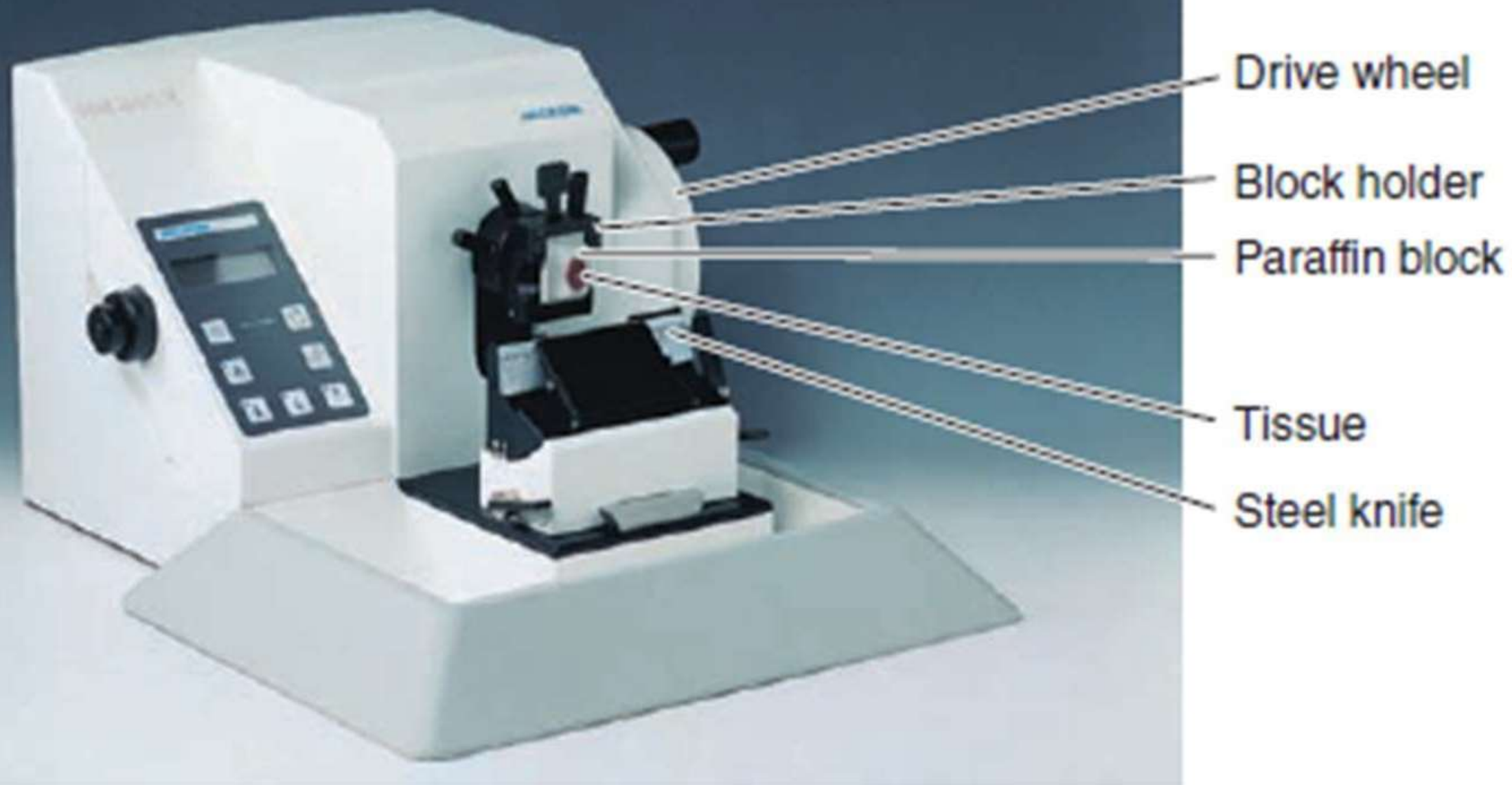
6-Sectioning:

Purpose: To obtain thin, uniform slices of the tissue for microscopic examination.

Method: The embedded tissue block is placed on a microtome, a cutting instrument, which produces thin sections

- ❑ A microtome is a device used for sectioning paraffin-embedded tissues for light microscopy.





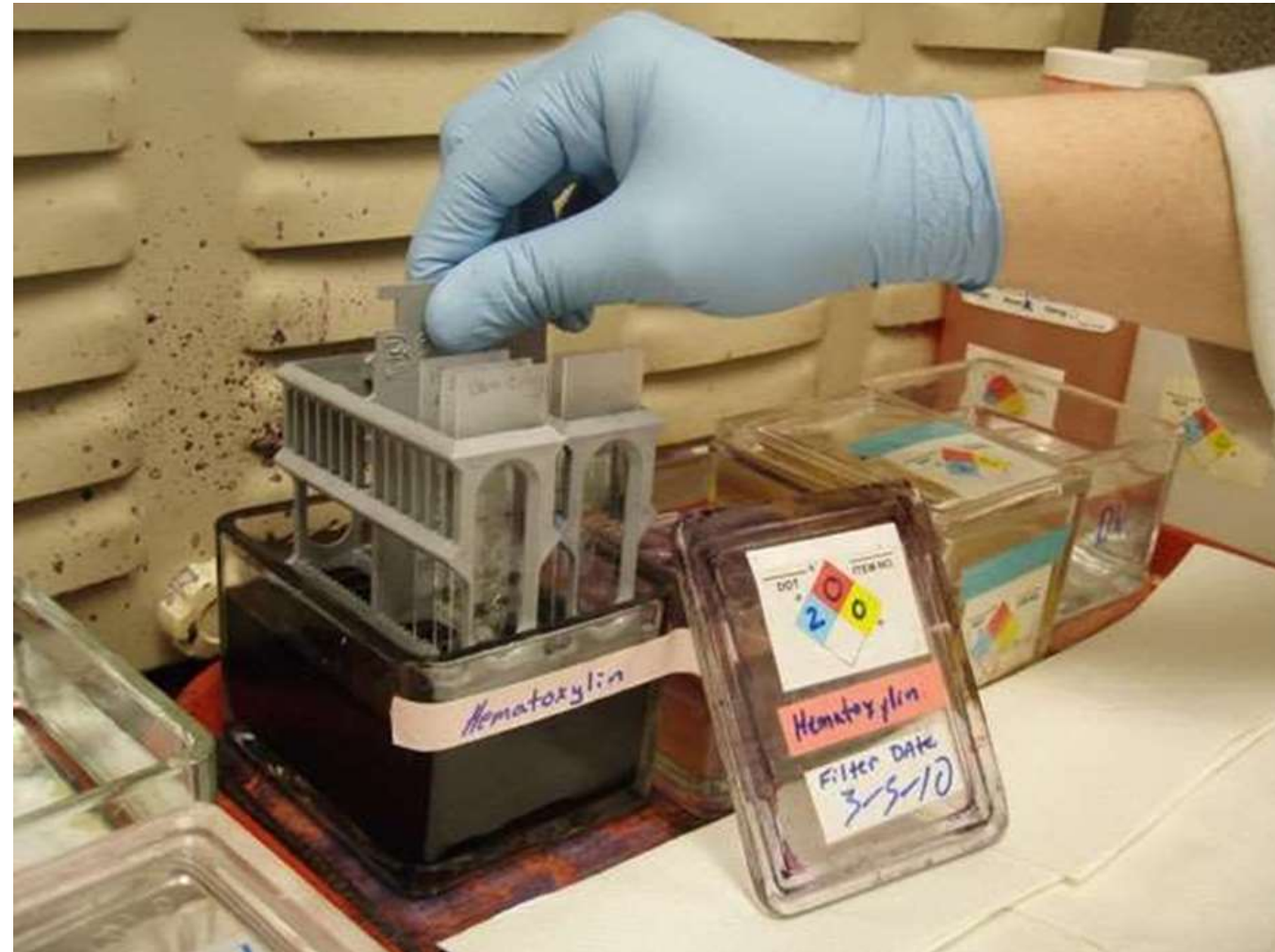
Steps in tissues preparation

7-Mounting and Staining:

Purpose: To enhance tissue contrast and highlight specific cellular components.

Method: Tissue sections are placed on microscope slides and stained most commonly with (H&E)

- ☐ Hematoxylin is a basic dye that stains acidic structures within cells.
- ☐ Eosin is an acidic dye that stains basic structures within cells.



Steps in tissues preparation

8-Microscopic Examination:

Purpose: To observe and analyze tissue sections under a microscope.

Method: Prepared slides are placed on a microscope stage, and different magnifications are used for examination.



